

Economics of the Adoption of Artificial Intelligence–Based Digital Technologies in Agriculture

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machine learning, precision farming, economic models, incentives

Abstract

Rapid advances and diffusion of artificial intelligence (AI) technologies have the potential to transform agriculture globally by improving measurement, prediction, and site-specific management on the farm, enabling autonomous equipment that is trained to mimic human behavior and developing recommendation systems designed to autonomously achieve various tasks. Here, we discuss the applications of AI-enabled technologies in agriculture, including those that are capable of on-farm reinforcement learning and key attributes that distinguish them from precision technologies currently available. We then describe various ways through which AI-driven technologies are likely to change the decision space for farmers and require changes to the theoretical and empirical economic models that seek to understand the incentives for their adoption. We conclude with a discussion of areas for future research on the economic, environmental, and equity implications of AI-enabled technology adoption for the agricultural sector.

1. INTRODUCTION

Technological innovations have greatly transformed agriculture over time and enabled it to triple crop yields since the mid-twentieth century, outpacing population growth and expansion of arable land, and to feed a growing population through the declining real price of food. The agricultural sector has experienced several technological revolutions over this period, starting with the Green Revolution in the 1950s that marked a transition to high-yielding varieties produced through plant hybridization that were more responsive to fertilizer and resistant to pests. This was followed by the Gene Revolution in the mid-1990s that introduced genetically modified crops via genetic engineering (Hamdan et al. 2022).

Continued technological development in agriculture has emerged in response to the increasing knowledge that agricultural landscapes are subject to significant variability in growing conditions due to variability in biophysical conditions such as soils, weather, and biological environment. This has led to the third agricultural revolution of precision farming enabled by the introduction of global positioning system (GPS) guidance for equipment and the ability to map spatial variability in growing conditions and vary input applications within fields in response to it. While these technologies generate vast amounts of geocoded, high-resolution big data, using these data to enable better decision making on crop management remains a challenge. This has created the demand for data analytics that can utilize spatially varying field data to improve understanding of the cause-and-effect relationships in crop management and enable site-specific decision making in real time. At the same time, with a growing scarcity of labor in agriculture, there is increasing interest in autonomous systems that can implement complex decisions with precision and timeliness. Other emerging challenges for agriculture include the need for better crop health monitoring, disease detection, and crop protection from pests and diseases. There are increasingly limited chemical options in the face of growing resistance of diseases, insects, and weeds to pesticides or health and environmental concerns with respect to pesticide use in general.

Smart farming is emerging as the fourth agricultural revolution that seeks to enable precision farming with artificial intelligence (AI) and autonomous technologies to address the challenges described above (Walter et al. 2017). A European Commission-appointed group of experts (Eur. Comm. 2018, p. 7) has defined AI as “systems designed by humans that, given a complex goal, act in the physical or digital world by perceiving their environment, interpreting the collected structured or unstructured data, reasoning on the knowledge derived from this data and deciding the best action(s) to take (according to predefined parameters) to achieve the given goal.” The science of AI is based on machine learning, which uses computer algorithms and analytics together with vast amounts of data to recognize patterns and build predictive models that can progressively get more accurate over time by learning from the data (Jarrahi et al. 2023). This ability for learning makes smart farming technologies different from previous generations of information systems.

Although the transition from precision farming to smart farming has been gradual as farming operations have become increasingly more knowledge intensive and reliant on information technologies and AI, it is possible to identify three key distinguishing features of smart farming. First is the development of an increasing range of information technologies to gather high-resolution data about spatially and temporally varying growing conditions, plant growth and response to environmental stressors, disease conditions, weed intensity, and nutrient deficiencies; this information can now be collected during the growing season. The second key feature is autonomy, which relies on machine reasoning and robotics to make site-specific interventions in the field throughout the growing season and enables field management in ways that are beyond the reach of human labor or existing agricultural equipment alone. The third feature is the development of machine learning techniques to analyze these complex and high-resolution data

from multiple sources using AI to improve predictive power for making recommendations and reducing the uncertainties in crop management.

Together, smart farming technologies are enabling better nonautonomous decision making by farmers and autonomous decision making using robots by using complex data and data analytics that involve machines self-learning from past errors and outcomes. These technologies are being employed to gather, process, and analyze farming data, automate processes to reduce labor time, monitor operations remotely, and digitally record operational decisions and activities. These technologies enable data interpretation, greater predictability and efficiency of farming operations, and real-time application of precision farming.

This review has several goals. First, we discuss ways in which AI-enabled technologies are being applied in agriculture and can enable a transformation in the way that data are used to generate actionable information and implement decisions with less human intervention (Javaid et al. 2023). Second, we review the ways in which technology adoption decisions are modeled and analyzed in the existing literature. We then describe various ways through which AI-driven technologies are likely to change the decision space for farmers and require changes to the economic models that seek to understand the incentives for adoption. Third, we investigate the approaches to empirically analyze the incentives for adoption of AI technologies. We discuss the extent to which the economic incentives for adoption of AI technologies are likely to be similar to those of existing technologies and learning from existing studies can be applied to understand the factors that will explain adoption of AI technologies. We also discuss the need to incorporate the attributes of AI-enabled technologies in empirical models analyzing the incentives for their adoption. We conclude with a brief discussion of areas for future research on the economic, environmental, and equity implications of adoption of AI-enabled technologies.

2. APPLICATIONS AND ATTRIBUTES OF ARTIFICIAL INTELLIGENCE-ENABLED TECHNOLOGIES IN AGRICULTURE

We classify the ways in which AI is being applied in agriculture into the following three categories and their potential economic and environmental benefits: (a) measurement, prediction, and site-specific management; (b) robotics and autonomous management; and (c) recommendation systems; we discuss each of these capabilities below (see **Table 1**). We then discuss some distinguishing features of AI-enabled technologies that have implications for the technology adoption decision.

2.1. Measurement, Prediction, and Site-Specific Management

An immediate value of AI is expected to be through providing precise, high-frequency, and high-spatial-resolution measurements about a broad range of field and crop conditions, such as crop health, nutrient deficiencies, soil quality, irrigation needs, and maturity of a crop for harvest (Liakos et al. 2018). New methods of image processing and species recognition are enabling automatic detection and classification of parasites, weeds, and diseases for real-time control. Machine vision with pattern recognition algorithms and automatic classification techniques can be used to autonomously monitor crop health, detect weeds, and distinguish between healthy and infected plants before outbreaks (Sharma et al. 2020, Mahlein et al. 2022, Günder et al. 2024). For example, machine learning algorithms can analyze images with leaf spectra to detect and discriminate between healthy plants and those that are infected by fungus with more than 95% accuracy (Pantazi et al. 2017).

The ability to better measure and monitor agricultural systems is being aided by increased understanding of why phenomena are occurring in farm systems. The potential to gather and analyze vast amounts of data with variability in crop genetics (G), environment (E), and management (M)

Table 1 Examples of AI-enabled applications and potential effects

Technology application	Functionality of the technology	Potential economic and environmental effects
Measurement/description	Provides precise, high-frequency, and high-spatial-resolution measurements of field and crop conditions Conducts image recognition of healthy versus infected plants; ML techniques distinguish crops and fruits in different maturation stages	Reduces labor and costs, increases efficiency, and improves crop yield
Site-specific management	AI inferences of soil moisture and weather forecasts can optimize targeted and timely irrigation decisions AI technology enables application of chemicals at the precise time and place by using AI to distinguish soil and crop needs and determine optimal responses	Increases the efficiency of input use, reduces input costs and labor requirements, improves soil quality, maintains adequate levels of nutrients, and increases crop yields and quality while reducing environmental contamination
Prediction	Predicts yields and forecasts potential harms by analyzing complex data metrics such as wind speed, extent of drought, pollination success, etc.	Yields better decision support systems and reduces waste
Robotics and autonomous management	Weeding robots can detect weeds by digital image processing and mechanically remove weeds Drones, unmanned aeronautical vehicles, or unmanned aircraft equipped with GPS and other sensors can monitor crop health, irrigation needs, and soil health and apply interventions autonomously	Extends crop management options that are not restricted by timing of intervention, improves farm efficiency, reduces the need for the human decision-making process, and improves environmental sustainability of farming operations
Recommendations	Evaluates different options for actions and provides information on advantages and disadvantages of different choices Provides support information on decisions when there are multiple objectives	Yields better decision support systems and recommendations of actions

Abbreviations: AI, artificial intelligence; GPS, global positioning system; ML, machine learning.

from millions of acres using machine learning is enabling a better understanding of the role for GxExM in explaining outcomes. Precision agriculture, together with machine learning and signal processing techniques, creates the ability to predict yields and forecast potential threats to a farming operation and enable timely responses. For example, AI approaches use image capture technology with the camera recognition tool or deep learning algorithms to conduct and identify possible defects and nutrient deficiencies in the soil (Ben Ayed & Hanana 2021). Descriptive field data can be used for prescriptive analytics to set up alert systems and automate recommended actions to respond to observed conditions. These site-specific management strategies using AI increase the efficiency of input use, reduce input costs and labor requirements, and increase crop yields and quality while reducing environmental contamination. However, these technologies can require higher upfront capital and learning costs, greater human capital skills, and more technical assistance.

2.2. Robotics and Autonomous Management

AI is enabling robotics, drones, sensors, and other automated systems that reduce the need for human decision making and implementation of tasks and improve farm efficiency and the environmental sustainability of farming operations. Robots are being developed to autonomously

plant crops, including cover crops; monitor crop health; detect and mechanically remove weeds; and spot and remedy the nutrient needs of plants. Drones or unmanned aerial vehicles (UAVs) or automated aircraft without a direct operator's control working in confluence with GPS and other sensors mounted on them are being implemented for crop health monitoring, irrigation equipment monitoring, weed identification, herd and wildlife monitoring, and disaster management (Talaviya et al. 2020).

As an example, cover cropping is being recommended to improve soil health and carbon sequestration by avoiding bare soil between harvest and replanting of crops such as corn and soybeans in the US Midwest. Conventional cover crop planting occurs after the harvesting of corn or soybeans when cool weather can hamper the establishment of the cover crop. The operation is also labor intensive and expensive. Small autonomous robots that can go underneath the canopy of the standing corn or soybean crop can plant cover crops early during the growing season without harming the cash crops and enable improved establishment of the cover crops with low labor intensity. Small semiautonomous robots are also being developed to detect and mechanically remove weeds by operating under the canopy of the crop, even during the growing season, reducing the need for chemical herbicides, reducing crop damage and slowing the growth of resistance (Khanna et al. 2022, Yu 2023).

Autonomous robots can extend the range of options for crop management that are not restricted by the timing of intervention or by their need for scarce labor (because they can now be undertaken during the growing season using small robots) and the number of hours of work in a day. The ability to use small robots could potentially make these technologies scale neutral and adoptable by farms of varying sizes, unlike large, indivisible equipment that is too costly for small farms to adopt.

2.3. Recommendation Systems

A challenge with high-spatial- and high-temporal-resolution big data being presented to a decision maker is that the decision maker cannot interpret the data or take action and becomes overwhelmed and ignores the data. This creates the need for interpretation, predictions, and recommendations to support decision making. AI systems are being developed to recommend actions a farmer can take by evaluating the different options for actions and the advantages and disadvantages of different choices. AI-enabled technologies can be designed to achieve one or more goals, can enable users to explore the different choices and their associated costs and benefits, and provide support for decisions that involve trade-offs. Increased data, predictive capabilities, and understanding of cause-and-effect relationships between actions and outcomes over different timescales open the possibility of managing the farm to meet multiple objectives. An emerging development in recommendation systems is that of active learning in which the algorithm determines the data it needs to make more accurate recommendations and enables a choice between cost and the desired level of accuracy. For example, machine learning could be used to identify which, where, and when moisture measurements should be made over a field based on a dynamic and spatial map of water availability (Smith 2020).

2.4. Development and Deployment of AI Technologies in Agriculture

Digital technologies vary in their extent of deployment and development; some technologies such as robot milking use GPS guidance for steering tractors, and fertilizer applicators and pesticide sprayers are used widely, while others such as UAVs are increasingly being adopted. Mobile autonomous equipment is still in the research and development stage (Lowenberg-DeBoer 2022). Weeding robots and cover crop planting robots are used in trials and becoming commercialized.

According to Erickson & Lowenberg-DeBoer (2021), 4% of agricultural input dealerships use robots for crop scouting services and 2% use them for providing weeding services. However, those dealers anticipate substantial growth by 2024, with 18% expecting to offer robotic crop scouting and 13% expecting to offer robotic weeding.

2.5. Attributes of AI-Enabled Technologies with Implications for the Adoption Decision

We consider three attributes of AI-enabled technologies that make them different from existing technologies and likely to be crucial in influencing adoption decisions.

2.5.1. Learning capability of AI technologies. AI allows a machine to simulate human intelligence to solve problems, which it can do through various ways such as machine learning. Machine learning can take various forms. Some of them involve training algorithms using historical input and output data to have the capability to determine patterns, identify species, make causal inferences, and generate predictions and recommendations. Other forms of learning, referred to as reinforcement learning (RL), involve machine learning by interacting with its environment. With RL, a technology learns to achieve a specified goal through trial and error and getting a positive or negative reward based on outcomes (see Gautron et al. 2022 for an introduction to RL in the context of crop management and Chen et al. 2021 for an example of RL for irrigation scheduling). RL allows for continued improvement of the technology after it is adopted, either nonautonomously through on-farm experimentation by the farmer or autonomously by the machine, with a lack of human supervision. This differs from other forms of machine learning where learning occurs exogenously to its use by the farmer. The possibilities for autonomous on-farm learning and experimentation during the use of the technology add new aspects that need to be considered for the adoption decision and are very different than those for existing technologies.

2.5.2. Black-box nature of AI decisions. Advanced AI algorithms are being trained to predict complex phenomena and generate recommendations whose basis may not be transparent or easy for farmers to fully understand. This black box nature of machine learning-based technologies also creates the possibility that a farmer may not fully understand the weaknesses of the algorithm and when it might fail. For example, an AI-enabled technology may be trained to operate under specified conditions, and changes in those conditions can affect the accuracy or lead to failure of AI systems (Sparrow et al. 2021). Thus, AI systems need to be frequently inspected, tested, and possibly retrained to prevent them from making the wrong recommendations. This imposes new risks and uncertainties for the decision maker; it also requires a farmer to relinquish control of decision making to an algorithm. Instead of a human interpreting the data and deciding, AI technologies rely on machines that can make the wrong decision on their own because they lack the sensory and cognitive capabilities of a human being.

2.5.3. AI-enabled technologies equipped with an objective function. AI-enabled technologies with autonomous RL require a clear definition of the objective function and the possibility to precisely measure the score achieved in each episode, e.g., the scores in a computer or board game, so that the technology can continually improve at achieving its objective. In an agricultural context, this might be substantially more complex, as farmers' objectives are likely to be complex and difficult to describe formally and might vary over time and context. Additionally, farmers may be driven not only to maximize profits but also to achieve nonpecuniary benefits (Howley 2015, Howley et al. 2015, Dominici et al. 2019). Using RL for optimizing crop management at a field level would, therefore, need to consider not only yield or profits but also risk and resource constraints on the farm or household level. Additionally, time preferences would need to be

considered; they can change over time and are more complex than a simple discount factor (Wuepper et al. 2023).

3. ECONOMIC MODELS OF TECHNOLOGY ADOPTION

There is a large literature on technology adoption stemming from the work of Griliches (1957) that focuses on analyzing how farmer characteristics and farmland heterogeneity affect adoption decisions (see, e.g., Khanna & Zilberman 1997, Zilberman et al. 1997, Khanna et al. 1999). Models explaining economic incentives for adopting new technologies have been largely based on the threshold model introduced by David (1975), which sought to explain why adoption is limited to some farmers. Farmers are assumed to make decisions to achieve an objective that is typically either profit or utility maximization over a given time horizon depending on their risk and time preferences. The model incorporates heterogeneity across farmers in one or more key characteristics (e.g., farm size, land quality, degree of risk aversion) and allows capital costs, input costs, and yield benefits from adoption to vary with the heterogeneous characteristics of the farmers but assumes homogeneity of land operated by a farmer. The model seeks to explain the threshold level of one or more of these heterogeneous factors that explain why farmers above a cut-off level of one of these factors will choose to make a different adoption decision than those below the cut-off level.

Applications most closely relevant to analyzing adoption of AI-based technologies include those examining incentives to adopt input-use efficiency-enhancing technologies and technologies that reduce yield damage from pests (Caswell & Zilberman 1986). These models represent a discrete-continuous choice problem that involves solving for the profit-maximizing input use with each of the two technologies first and then comparing the maximized profits to predict the optimal technology choice at various soil quality levels. Khanna et al. (2002) extend this by incorporating the relationship between input-use efficiency and pollution generation from the field; inputs not utilized effectively are wasted and contribute to agricultural pollution.

This framework is further extended to account for in-field spatial variability in soil conditions and to analyze the decision by risk-averse farmers to adopt precision technologies for varying the rate of nitrogen application within a field instead of applying it at a uniform rate based on average soil conditions (Isik & Khanna 2003). Isik & Khanna (2003) incorporate the impact of these technologies on both returns and risks and show that uncertainty about soil conditions can lead risk-averse farmers to apply more fertilizer and generate more pollution than in the certainty case, even with precision technologies, and reduce the net benefits of adopting precision technologies. Their findings imply that AI-based technologies that improve the accuracy of precision technologies and reduce uncertainty about outcomes can increase the incentives for adoption and reduce fertilizer applications, particularly in fields characterized by greater spatial variability in soil conditions.

The threshold model indicates that new technologies are more likely to be first adopted in locations where the returns are highest and farmers are less risk averse and more willing to experiment with the technology. Over time, adoption rates could grow as the capital costs of the technology decline and as learning costs decline due to learning by doing (in the production of the product), learning by using (the technology oneself), and network externalities (learning from others using a technology).

In many cases, a technology is not embodied in a single piece of equipment but consists of a suite of complementary but optional components that serve different functions (Khanna 2001). These component technologies differ in their capital costs of adoption and the net returns from their adoption; returns from variable rate application technologies depend on the outcomes from

the adoption of other components, such as technologies that diagnose the extent of spatial variability within the field. As a result, some components may be adopted more than others, and components may be adopted sequentially, rather than as a package. This may also be the case with AI-enabled technologies that have many components or functionalities that can be adopted sequentially, necessitating the need for modeling adoption decisions at a more granular level.

Several studies have gone beyond analyzing the static decision to adopt/not adopt to analyzing the timing of adoption of a new technology (see Khanna et al. 2000). In cases where the returns and costs of a new technology are stochastic, adopters face a choice of when to adopt in the face of uncertainty and the possibility that costs of a technology may decline over time and/or the returns from adoption may increase over time. In these cases, studies have gone beyond a net present value-based framework applying a real option approach to show that adoption of technologies with high and irreversible sunk costs occurs when the marginal benefit overcomes marginal cost plus the hurdle rates that increase with uncertainty (Dixit & Pindyck 1994). The potential adopter must weigh two types of costs against each other: the cost of adopting too soon (because the decision is irreversible and at least part of the costs cannot be recovered) and the opportunity cost of waiting too long and forgoing potential payoffs in the meantime. In the context of precision technologies, Khanna et al. (2000) show that the value of waiting to adopt increases with uncertainty about output prices and expectations of declining fixed costs but decreases with spatial variability in soil conditions. Adoption is likely to occur sooner on fields with higher soil quality and fertility and greater variability in these conditions. Uncertainty can also create a preference for custom hiring rather than owner purchase of equipment that has a high likelihood of rapid obsolescence, reduction in cost, and performance improvements. The explicit consideration of the value of waiting implies that the timing of adoption also depends on how fast and by how much technology will advance over time. The process of technical innovation is inherently stochastic; thus, there is uncertainty about the speed with which new technologies become available for adoption and about the extent of gains in efficiency with the new technology relative to the current state of the art. These uncertainties may be comparatively large for AI-enabled technology given their current dynamic development and may favor rental of equipment or its services that reduces the cost of disadoption compared to owner purchase.

In many cases, farmers may adopt technologies partially or gradually in a stepwise manner, and defining adoption as a binary decision can be misleading in assessing its economic and environmental benefits (Montes de Oca Munguia et al. 2021). The process of adoption typically consists of six stages: awareness, nontrial evaluation, trial evaluation, adoption, review and modification, and disadoption (Pannell & Claassen 2020). Farmer characteristics such as degree of risk aversion and extent to which they are forward looking can affect the extent and timing of partial adoption. A technology perceived as highly risky will not be adopted on all acres immediately; instead, it may be adopted gradually as more information flows that changes the farmer's perception about the technology. Partial adoption improves knowledge about the new technology's profit distribution through self-learning. Farmers may be willing to adopt a new technology even if it is less profitable than no adoption in the current period because it will be more profitable in the longer run. Several studies have modeled the effects of self-learning and of learning from neighbors (e.g., Foster & Rosenzweig 1995, Baerenklau 2005, Ma & Shi 2015) on adoption decisions. They show that imperfect knowledge of the new technology and risk preferences can constrain adoption but that farmers who are forward looking see the potential for self-learning and peer group learning through partial adoption. AI-enabled technologies offer new ways to reduce the uncertainty of agricultural operations by using experimental trials to improve the predictive power and recommendations of machine learning technologies, as described below.

Table 2 Choice of technology features and corresponding decisions

Execution	Recommendation		
	Not AI-enabled (1)	AI-enabled, without on-farm RL (2)	AI-enabled, with on-farm RL (3)
Not AI-enabled (A)	Farmer decides when to weed and uses a tractor to do so	Include image recognition for weed detection to initiate a previously determined spraying rule	Include image recognition for weed detection to initiate dynamic spraying rules based on feedback from previous actions and outcomes (e.g., weed count)
AI-enabled, without on-farm RL (B)	Farmer makes the decision to weed and conducts automated robotic weeding based on previously determined spraying rules	Include image recognition for weed detection to initiate a previously determined spraying rule where execution uses AI-enabled field navigation	Decision-making rules are improved based on feedback from previous actions and outcomes where execution uses AI-enabled field navigation
AI-enabled, with on-farm RL (C)	Farmer makes the decision to weed and releases weeding robots whose navigation improves based on feedback from previous actions and outcomes	Include image recognition for weed detection to initiate a previously determined spraying rule where execution is improved using feedback from previous actions and outcomes	Decision-making rules and execution are improved based on feedback from previous actions and outcomes

Abbreviations: AI, artificial intelligence; RL, reinforcement learning.

4. NEW ECONOMIC MODELS OF ARTIFICIAL INTELLIGENCE TECHNOLOGY ADOPTION

The decision to adopt AI technologies that mainly aim to improve sensing and monitoring capability and provide temporal and spatial information with higher resolution is likely to be determined by very similar behavioral processes as those to adopt existing precision farming technology. On the other hand, AI technologies that use RL to make recommendations and execute them may fundamentally expand the types of decisions farmers will be faced with when choosing among technologies. We therefore distinguish between three types of technology choices that a farmer is likely to face: (a) not AI-enabled, (b) AI-enabled without on-farm RL, and (c) AI-enabled with on-farm RL. We focus on two of the applications in **Table 1**, namely recommendation and execution of management, and illustrate the example of the choice of technology for recommending weeding and executing the weeding decision (see **Table 2**).

On one end of the spectrum in **Table 2**, farmers make their decision and execute it without any AI support. This is the case, for example, of a farmer who decides to weed based on manual inspection or a calendar basis and uses some non-AI-enabled technology such as a tractor to weed (cell A1, **Table 2**). The farmer might instead use an AI-enabled technology without on-farm RL such as an image recognition software for weed detection to initiate weeding and execute it using the same tractor (cell A2). While still using the same tractor and image recognition software, an AI system may recommend spraying rules that get better over time based on RL feedback from sets of previous actions (e.g., time of weeding) and outcomes (e.g., weed count) (cell A3).

Similarly, a farmer might replace the tractor for execution by technologies that are AI-enabled that might have used RL during the production process but that do not allow continuous on-farm learning and improvement (row B) and AI-enabled with on-farm RL (row C). One example of weeding execution using an AI-enabled technology without on-farm learning is the use of semi-autonomous small robots to weed where the robot uses AI technology to autonomously navigate a field (cell B1) (e.g., Yu et al. 2024). Alternatively, robot navigation ability can use on-farm RL to

use feedback from previous actions (e.g., navigation route) and outcomes (e.g., number of times robot tips over) to improve navigation over time (cell C1).

As with most technologies, those along that spectrum differ in the expected costs and benefits relative to the status quo technologies (as described in **Table 1**). In addition, as we move from cell A1 to cell C3 in **Table 2**, a farmer's adoption decision involves a larger decision space, a new type of endogenous risk, and a new source of aversion to new technologies. The decision space is larger because AI-enabled technologies allow decision making and execution to occur at smaller time steps and spatial units within a growing season (cell B2). As a result, a crop management plan developed at the start of the growing season could ideally be adapted to events that occur during the growing season. This requires information about the crop and field conditions during the growing season, knowledge about the management response, and the technology that enables the implementation of that response with a standing crop. AI methods can enable the technology to learn the unique site-specific management needed for a particular farm through on-farm experiments implemented on that farm.

The adoption decision now also involves additional risk because AI-enabled technologies might lead to bad outcomes if farming conditions differ from those used to train the AI. That risk can be reduced in the case of technologies that allow on-farm RL in recommendation formulation and execution (cell C3) because farmers have the option to make two additional decisions: how much land and human capital to allocate for on-farm RL and whether to share data with technology providers and farmers so that on-farm RL-based recommendations and executions can be continuously improved based on RL happening elsewhere. We describe these decisions below.

AI-enabled navigation and AI-enabled decision making (cell B2) based on real-time data increase the decision variable space for farmers. For example, compared to weeding using a large tractor (e.g., weeding in corn systems), the decision is more complex with small AI-enabled robots that can operate autonomously underneath the canopy of a standing crop. Farmers now face new decisions, such as the desired autonomy level of the AI-based technology, the optimal timing and frequency of deploying robots during the growing season to kill cohorts of weeds, the number of robots to deploy, and where to deploy robots depending on the varying spatial pattern of weed emergence.

The resulting increased complexity of farm management decisions requires farm optimization models with a high spatial and temporal resolution to analyze the optimality of farmer decision making over time and space. While such models exist (e.g., Atallah et al. 2015), they have so far had a limited application due to the labor intensity of high-spatial-resolution scouting and the knowledge required for pest and disease identification. In addition, while existing models incorporate RL in the recommendation, such as the adoption of a pest threshold to spray a crop (e.g., Siddique 2019, Siddique et al. 2019) or the adoption of a water-saving technology (e.g., Vinyals et al. 2023), these models do not incorporate on-farm RL that improves performance and outcomes after adoption and throughout the growing seasons.

In addition to involving a larger decision space, AI-enabled technologies with mixed on-farm RL regarding execution and recommendation (cell B2) involve a new type of risk of failure. This risk stems from a potential mismatch between historical data used to train the AI technology and the environmental conditions under which these occur. This risk of failure is exacerbated in the case where both recommendation and execution are AI-enabled. For instance, a technology might be trained to operate in certain soil types, soil moisture, light, and temperature conditions that are different from the realized farm conditions the day an agricultural practice is implemented with such technology (Magistri et al. 2023). At the same time, the image recognition the technology uses for species identification might be based on datasets that do not include the species encountered, leading to a risk of failure in both recommendation and execution.

Farmers adopting a technology that uses on-farm RL (cell C3) are then faced with an additional decision to invest in learning for their technology by allocating land and human resources for experimentation when field conditions, crops, and biotic and abiotic factors affecting them differ from those under which a robot's system was trained. As such, the risk of failure due to a mismatch between the source domain of the training and the application domain becomes partially endogenous to farmers. Farmers are likely to be heterogeneous in their willingness and ability to invest in on-farm RL to reduce this type of risk. This is likely to be a driver of the heterogeneity in adoption and its success upon adoption and, therefore, in the likelihood of technological disadoption. Farmers will be faced with the decision on the threshold difference between their conditions and those of the source domain beyond which they would not adopt the technology at all and below which they would adopt the technology with or without investing in on-farm RL. This is a complex decision, and it is not immediately clear whether farmers have information on the environmental differences between source domain and own conditions and how they would make such a multivariate threshold decision.

If farmers decide to adopt a technology with on-farm RL (e.g., cell C3) and dedicate land and human capital for the learning process, then their adoption decision also involves a related decision to share (or not) data on the performance of robots on their farms with other farmer users and the technology developer. This additional decision to share data is especially important for technologies that generate RL-based farming recommendations because feedback based on yield realizations is slow (with one growing season in a year), and consequently, the learning process is slow unless farmers decide to share data from large numbers of fields.

Farmers are likely to display more aversion to technologies that provide AI-enabled and RL-based recommendations (columns 2 and 3 of **Table 2**) than technologies that use AI and RL for execution (rows B and C of **Table 2**). Although some farmers are used to the partial automation of farming equipment, they are likely to be averse to taking recommendations from an algorithm perceived as a black box (see Section 5), which presents a large departure from taking recommendations from farm advisors.

One justifiable reason for this aversion is that AI technologies with RL often have a single reward function that might or might not correspond to that of a farmer adopting an AI-enabled technology. The technology may generate optimal recommendations that differ from what are considered optimal by the farmer. Farmers may adopt but decide not to follow the recommendation by the technology and instead use heuristics to modify it based on other considerations. In case the discrepancy in objectives was not known before adoption, this can lead to outcomes different from those expected by a farmer when they adopted the technology. For an economic modeler, where such discrepancy was known before adoption, the prespecified RL reward function becomes a technological constraint in a farmer's economic objective function used to evaluate the adoption decision. In the future, it is possible that an AI technology would offer customizable recommendations based on several multiobjective RL algorithms representing different sets of multiple objectives farmers might consider. Currently, multiobjective RL is an active area of research, and most existing RL algorithms used in crop management support have a single reward function (Gautron et al. 2022).

Technology adoption models that are most appropriate to capturing the larger decision space include spatial-dynamic, integrated socio-ecological models. Given the potential of AI technologies to provide the possibility to conduct management actions through the growing season and customize them at the plant level, spatial-dynamic optimization models in the field of renewable resource economics are most relevant for modeling farmer decision making regarding the adoption of AI technologies. Until the last two decades, that literature was mostly nonspatial (see review by Wilen 2007). Later research showed how ignoring the spatial nature of biological

processes and spatial gradients of costs and revenues can lead to suboptimal decisions (Sanchirico & Wilen 1999, 2005). Authors have incorporated space in resource economic models by specifying location-dependent, state transition probabilities (e.g., Rich & Winter-Nelson 2007) or by incorporating spatial variation in population dynamics (Holmes et al. 1994). In such models, spatial heterogeneity is static (see the review by Smith et al. 2009), whereas it can be dynamic and endogenously determined by the implementation of AI-based and RL-enabled strategies.

In contrast, cellular automata and agent-based models (ABMs), also called individual-based models (IBMs) in the ecology literature (Grimm & Railsback 2005), allow study of a population of heterogeneous agents living in a spatial-dynamic world where agent types can range from agents without agency such as plants (e.g., Atallah et al. 2018, Wu et al. 2023), agents with algorithmic learning such as robots (e.g., Kuandykov et al. 2015), and agents with social learning such as farmers (Cardona Santos et al. 2021). ABMs are computationally intensive simulation models where agents interact with their environment and each other according to rule-based algorithms and mathematical equations (Tsfatsion 2006, Shang et al. 2021). The ABM literature includes applications to socio-ecological resilience (Grêt-Regamey et al. 2019), policy analysis (Granco et al. 2019), fishery economics (BenDor et al. 2009), industrial organization (Boyer & Brorsen 2013), production economics (Akanle & Zhang 2008), and natural resource management (Rasch et al. 2017). Applications in agricultural technology adoption, however, are less common (but see Berger 2001, Schreinemachers et al. 2007, and Vinyals et al. 2023 for exceptions and Shang et al. 2021 for a recent review of the empirical literature on the adoption of digital farming technologies and a conceptual framework for ABMs of technology diffusion).

ABMs are particularly appropriate for modeling how a biophysical process spreads across plants, how robots navigate and analyze information on those plants, how a group of robots would interact with one another to optimize the task at hand, how adoption by farmers might influence adoption by their neighbors or members in their social networks, and how a group of farmers might engage in collective action through data sharing that improved their RL-based technologies. An ABM can be constructed from the point of view of a farmer optimizing adoption (Atallah et al. 2015, Wu et al. 2023) or one of a policymaker wanting to analyze adoption across multiple farmers over time and space, using knowledge and assumptions on the determinants of farmer adoption (Vinyals et al. 2023). ABMs can be embedded in one another so that an ABM of plant agents can be embedded in an ABM of farmer agents (Atallah et al. 2017). To find an optimal solution, an ABM can simulate the deployment of technologies in a heterogeneous population of crops and generate distributions of bioeconomic outcomes under alternative spatially and temporally targeted technology deployment strategies and investments in on-farm RL that can be ranked to obtain spatially and dynamically optimal allocation of AI technologies on the farm. Outputs can then be statistically analyzed to evaluate alternative strategies or policies (Fagiolo et al. 2007).

5. EMPIRICAL DRIVERS OF TECHNOLOGY ADOPTION

A large body of empirical literature seeks to examine, *ex post*, the economic, noneconomic, and behavioral drivers of adoption of a new technology, including precision technologies (Prokopy et al. 2008, Streletskaia et al. 2020, Weersink and Fulton 2020). Finger et al. (2019), Khanna (2021), and Shang et al. (2021) provide recent reviews of the empirical literature on the determinants of adoption of digital technologies. The extent to which the findings of these studies provide useful insights to predict adoption patterns of novel AI technologies depends on the extent to which we expect technology attributes and farmer behavior to overlap between existing technologies and AI technologies. Precision farming technologies share many of the characteristics of technologies in

rows A and B and columns 1 and 2 of **Table 2**, while the aspect of learning (on the farm) in cell C3 adds novel aspects of the technology that are likely to impact the factors that will affect adoption decisions.

Additionally, as discussed above, the type of adoption decision might also become more complicated because AI technologies add additional dimensions, such as the level of autonomy or the degree of on-farm learning a farmer allows and whether that is in decision making and/or execution of farm operations. These aspects are most relevant for AI technologies that aim to automate management or that provide management recommendations (see examples in rows 4–5 in **Table 1**). For AI technologies that mainly aim to provide relevant information but where management decisions are left with the farmer (rows 1–3 in **Table 1**), the determinants of adoption decision might be more similar to those for existing (precision) farming technologies. Most economic analyses of technology adoption assume that adoption decisions are instantaneous and guided by the objective of profit maximization. However, as noted by Weersink & Fulton (2020), it would be more realistic, particularly with emerging technologies such as AI technologies, to undertake ex ante analyses and to recognize that adoption is a multistage process, including awareness, nontrial evaluation, trial evaluation (experimentation), adoption, and revision (reevaluation), and could be followed by disadoption.

Following Shang et al. (2021), we group the explanatory factors into (a) farm characteristics, (b) interactions, (c) institutions, (d) attributes of technology, (e) psychological factors, and (f) operator characteristics. We discuss the extent to which these factors are likely to explain AI technology adoption and in which areas we expect differences compared to previous technologies.

5.1. Farm Characteristics

Farm characteristics are among the most studied and most important factors that determine digital technology adoption. These include farm size, which is typically found to positively relate to adoption, with larger farms being able to leverage economies of scale and spread fixed investment cost more widely. These factors are likely to remain equally relevant for AI technologies. A case where AI technologies might be different is if swarm robotic systems become more relevant. In those systems, the problem of indivisibility might be lower, and managing larger swarms of robots could lead to diseconomy of scale effects in case management complexity increase with larger swarms. Novel robotic systems might also change the strength of scale effects, for example, using cost calculation based on experiments from an experimentation farm. Lowenberg-DeBoer et al. (2022) show that the economies of scale effect might be lower for robotic technologies than for conventional technologies.

5.2. Interactions

Interaction between peers, the influence of extension, farmer organization, or technology providers might be equally relevant for AI technologies as for non-AI technologies. One aspect where AI technologies might be fundamentally different than existing technologies is in its data network effects (Gregory et al. 2021). Typically, AI and machine learning systems become better when more training data are available. This might lead to data network effects where the usefulness of the entire system increases with each additional user. For example, the rideshare company Uber uses machine learning algorithms to analyze data collected from each user and to improve matching and experience offered to other riders and users (Gregory et al. 2021). Similarly, an open AI ecosystem is expected to improve healthcare services to customers by using the unprecedented amount of data available combined with advances in natural language processing and social awareness algorithms (Ludik 2016).

Following Gregory et al. (2022), the condition for data network effects to appear is that (a) learning from one user's data can be translated into a better product of another user and (b) this improvement is achieved quickly, such that it improves the current performance of the product (Gregory et al. 2022). These effects are described for nonagricultural AI technologies (Gregory et al. 2021). Such network effects could lead to more monopolistic suppliers of a technology, create a need for data sharing agreements, and raise the importance of data ownership. This feature of AI technologies could create dynamically varying incentives for adoption because the value of adoption could be relatively low in early stages (when minimal training data are available) but grow with increasing adoption by peers (as more training data become available). This could create an incentive to delay adoption and wait until more training data are available. Additionally, the value of the technology for earlier adopters would change over time if the performance of their equipment improves with increasing adoption by other farmers. This could create an incentive for early adopters to actively promote adoption by others, increasing the importance of peer effects. This is a distinct aspect of AI technologies, because with non-AI technologies, the capability of technology is typically relatively independent of the usage of others.

5.3. Institutions

Shang et al. (2021) identify access to subsidies and credit as well as laws and regulations as important institutional factors determining adoption of precision farming technology. Those factors seem equally relevant for AI-based technologies. Lowenberg-DeBoer et al. (2022) identify three main regulatory issues that are crucial for adoption of autonomous agricultural equipment: (a) regulation for onsite supervision, (b) liability for machine errors, and (c) intellectual property in robotic learning. Institutions covering data access, sharing, and usage are expected to become even more important for AI technologies and in order to realize data network effects. Legal and technical standards are important to avoid data lock-in in machines and devices that limit competition to develop new systems benefiting from large datasets collected on many farms (Atik & Martens 2021).

5.4. Technology Attributes

Previous research shows that the trialability of a technology can affect the decision to adopt it. Trials provide information about the suitability of a new technology and reduce uncertainty about its net benefits, which can be particularly important for risk-averse farmers. Small-scale experimentation on the farm (Bullock et al. 2019) can reduce the risk of failure. With non-AI technologies, Weersink & Fulton (2020) define trialability of a technology as being positively related to its simplicity, divisibility, and observability and inversely related to the cost of undertaking a trial.

Trialability is likely to be important with AI-enabled technologies but for different reasons. As discussed above, a crucial feature of AI-enabled technologies with learning is the need for trials to train the technology; trials enable the technology to explore and learn by trying out new actions in order to observe their outcome and improve the performance of the technology. Here, trials are critical for training the technology and most successful in settings where the algorithm can try out different actions in millions of repetitions, for example, in a large number of self-plays when using RL learning to play computer or board games (Mnih et al. 2015). Gautron et al. (2022, p. 16) found that between 500,000 and 1,000,000 episodes are used during training in existing applications of RL in crop management, where one episode would be one cropping year in the real world. Clearly, this number of episodes can only be reached with *in silico* simulations (e.g., using crop growth simulation models), assuming these models themselves have high accuracy and prediction power.

Crop simulation and pest dynamics models can generate data to train RL algorithms. In that sense, such data can be substitutes for empirical observations, especially when the frequency of natural events of interest such as outbreaks is low for timely training or if the time horizon of interest is longer than that of existing data. For instance, while Meisner et al. (2016) use observational data to provide machine learning–based pest management recommendations (last row in **Table 1** and columns 1–3 in **Table 2**), Siddique et al. (2019) achieve the same objective using a bioeconomic simulation model as a data-generation process. Crop simulation and pest dynamics models can, therefore, play an integral part in on-farm learning. In practice, however, these models are extremely large and require many data for calibration before they can be used for locally relevant simulated data, and they are all very specialized in one dimension (e.g., soil carbon and nitrogen cycles) and either simplify or ignore other dimensions (e.g., crop growth and crop–pathogen interaction). As a result, while these models can help improve the performance of a technology, they can also increase the risk of failure in the process of learning. These models are also designed to be used by researchers and not designed to interface with farmers doing on-farm trials. Finally, the role of such models is limited to the AI-enabled farm recommendations (columns 1–3 in **Table 2**) and not AI-enabled execution of farming tasks (rows A–C in **Table 2**).

From an adoption perspective, it is questionable to what extent farmers might accept this experimentation on their farm, which inevitably might lead to crop failure in some trials. Hence, RL-based technology adoption does not only come with the typical risk of adopting a new technology, but applying the technology brings an inherent uncertainty through the need for exploration required for learning. Hence, adoption studies need to put a particular focus on risk and loss preference of farmers, which might determine how much exploration a farmer is willing to allow during training; risk aversion may limit the risk of failure but also slow down learning, potentially limiting the positive benefits of the technology.

5.5. Psychological Factors

Another crucial feature of AI-based decision making or recommendations is that they might be perceived as a black box and lead to a psychological aversion to adoption. The rationale for the recommendations of AI (or RL) systems may be difficult to understand and lead to farmer unwillingness to accept when making important crop management decisions. Trust in the AI technology as well as the provider of that technology is likely to be a crucial determinant of adoption (Ku & Wang 2022). This has also been noted in the context of AI applications in finance, which can expose users to risks due to biases in the training data, erroneous data, or erroneous preprocessing and reliance on poorly performing models (Cohen et al. 2021). Farmers who are risk averse are likely to be unwilling to adopt AI technology if it exposes them to data and model risks. Another concern is algorithm aversion, which describes the preference for a human decision over an AI-based decision, even if the AI-based system performs better on a given task. This has been discussed in legal, medical, and financial settings (Mahmud et al. 2022, Chang & Wang 2023). Niszczoła & Kaszás (2020) find that individual investors are averse to investment decisions made by so-called robo investment managers. The degree of algorithm aversion varies with individual characteristics (Longoni et al. 2019), and ways to mitigate them are being actively debated (Longoni et al. 2019, Pezzo & Beckstead 2020). This has implications for understanding the extent to which farmers can be expected to show algorithm aversion: the drivers of this aversion and the heterogeneity in this aversion across farmers that differ in their attitudinal preferences.

Uncertainties related to the benefits of the technology and high learning cost are likely to favor custom hiring over owner purchase of the technology. However, this is likely to exacerbate the black box nature of the adoption decision and exacerbate concerns about loss of control for

certain types of farmers. Differences between the objectives of the farmer and those that a robot may be trained to achieve and between the acceptable trade-offs each is willing or trained to make can lead to differential incentives for adoption across large and small farmers, owners, and renters or across other farm categorizations. Thus, the design of the technology in terms of its potential to allow user customization of objective functions embedded in RL algorithms can affect who adopts the technology.

5.6. Operator Characteristics

Shang et al. (2021) find that operator characteristics such as age, income, computer use, farming as a main occupation, off-farm income, farming experience, innovativeness, knowledge, and risk preferences affect adoption behavior, although the effects vary with technology and across studies. Many of these factors are likely to be equally relevant for AI technology, although the relative importance of risk preferences, innovativeness, and learning costs may be greater for AI-based technologies. As discussed above, RL requires a clear and formal definition of the objective function and the possibility to precisely measure the score achieved in each episode. In an agricultural context, this might be complex, as farmers' objectives are likely to be complex and difficult to describe formally, and might vary over time and context. For the design and development of an AI system, this creates a challenge, as those aspects would need to be known and quantified precisely. This would be particularly challenging for nonpecuniary benefits that would be difficult to quantify even for the farmers themselves. From an adoption perspective, we can expect that the importance of different nonpecuniary benefits might vary across different types of farms (commercial versus hobby, part-time versus full-time) or individual farmer characteristics (different norms and beliefs). Hence, farmers that have less-complex objectives (i.e., more one-dimensional, profit-oriented) are more likely to adopt because those aspects are easier to quantify in the AI system.

6. CONCLUSIONS

Agriculture is poised to undergo a substantial transformation with the diffusion of AI-enabled technologies; the extent and form that this will take will depend on which technologies are adopted and the path and speed of diffusion in different settings across the world. For economists to contribute to a better understanding of the determinants of AI technology adoption and thereby its consequences for the profitability and environmental sustainability of farming, it is important to understand the novel attributes of these technologies and the new types of decisions farmers will face when adopting them. We described some key distinguishing attributes of AI technologies and discussed their implications for economic modeling of the drivers of the adoption decision.

The ultimate capabilities of AI and especially AI technologies with RL will depend on the data that are being used and by whom they can be used. Whether farmers are willing to conduct on-farm experimentation to allow the machines to learn depends on the return for this costly activity. This in turn depends on the customized knowledge already embodied at the time of adoption and thereby on the maturity of the technology. The speed to maturity will also depend on farmers' willingness to share the data with technology providers and other farmers and therefore on provisions and regulations related to data security and rights of use.

We now briefly discuss some economic, equity, and environmental implications of the adoption of AI technologies that are relevant for future economic analysis of AI technologies. First, AI-enabled and autonomous technologies may alleviate some and aggravate other equity-related issues in agriculture. On the one hand, autonomous technologies can improve the working conditions of farmers, relieving them of manual, routine tasks, and the health hazards they pose, and

the divisibility of small robot and drone technologies can make them suitable for smaller farmers. On the other hand, the capital and learning costs of AI technologies will likely favor the adoption by big farms initially. Moreover, poor Internet penetration rates, access to AI technologies, and an unwillingness to engage with digital technologies, even if they are available, can create a digital divide that can widen the inequities between commercial farmers and smallholders (Hennessy et al. 2016).

Second, widespread reliance on AI technologies exposes the agricultural sector to new risks, including cyberattacks and external interference with AI-driven machinery, such as autonomous tractors and harvesters, robot swarms, and autonomous sprayers. Rapid diffusion of autonomously learning machines can also lead to accidental risks, where large-scale reliance on the same suite of algorithms can lead to model error or poorly calibrated sensors to lead to excessive fertilization, soil health degradation, and risk of large-scale crop yield failures (Tzachor et al. 2022).

Lastly, AI-enabled agricultural systems have the potential to be more environmentally friendly because they can reduce chemical applications, improve soil and crop health, increase input-use efficiency in agriculture, and substitute away from chemical inputs while increasing productivity (see Finger 2023). The extent to which this will be the case is uncertain and will depend on the accuracy and predictive capability of the technology; inaccurate recommendations could also lead to widespread harm and unsustainable practices. An additional benefit of AI-enabled technologies could be their potential to enable voluntary and regulatory efforts to reduce agricultural pollution. The nonpoint nature of agricultural pollution has precluded the development of performance-based policies that can be implemented at low information costs to reward/penalize farmers based on the pollution outcomes resulting from their agricultural production efforts. AI-enabled technologies offer new possibilities for monitoring and documenting management practices and attributing environmental outcomes to those practices. This can enable low-cost implementation of results-based ecosystem payment schemes that can achieve environmental targets more cost-effectively than existing uniform practice-based payments. We leave further discussion of these issues for future research.

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